



Aerobic oxidation of alkylaromatic hydrocarbons to hydroperoxides catalysed by *N*-hydroxyimides in ionic liquids as solvents

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ABSTRACT

This paper reports studies on the oxidation of alkylaromatic hydrocarbons to hydroperoxides with oxygen in the presence of *N*-hydroxyimides (*N*-hydroxyphthalimide, *N*-hydroxysuccinimide, 4-dodecyloxycarbonyl-*N*-hydroxyphthalimide, *N,N*-dihydroxypyromellitimide, *N*-acetoxyphthalimide) in ionic liquids as solvents. Cumene, ethylbenzene, mono- and di-isopropyl derivatives of naphthalene and biphenyl were used as starting materials. As solvents, ionic liquids composed of 1-alkyl-3-methylimidazolium cations with ethyl, butyl, hexyl or octyl substituents and anions such as [BF₄], [PF₆], [NTf₂], [CF₃SO₃], [O₂SO₃], and [CH₃OSO₃] were applied. It has been established that organic solvents could be replaced by ionic liquids, which does not accelerate hydroperoxide decomposition and is characterized by relatively high oxygen and catalyst solubility and low viscosity. Among studied ILs, these with [NTf₂] and [PF₆] anions meet the above requirements.

1. Introduction

Processes of alkyl aromatic hydrocarbons oxidation with oxygen to hydroperoxides are used on an industrial scale to obtain specialized chemical substances. The most important hydroperoxide, related to the scope of production, is cumene hydroperoxide, which is produced through oxidation of cumene and then used as a semi-product for the synthesis of phenol via the Hock method (Fig. 1). It is used to a lesser degree as an oxidizing agent in a reaction of propylene epoxidation to propylene oxide in a process named 'PO only' [1,2]. In the industry, cumene is oxidized with oxygen or air at a temperature ranging from 100 to 140 °C under pressure of approximately 0.6–0.7 MPa. Low degrees of conversion equal to 20% and selectivity to hydroperoxide of 90–95% are achieved.

An important role in industrial synthesis is additionally played by ethylbenzene hydroperoxide, used chiefly as an oxidizing agent in propylene oxidation to propylene oxide. Oxidation of ethylbenzene to its hydroperoxide is performed at a temperature of 140–150 °C, under pressure of 0.3–0.4 MPa, with conversion of 7–8% and maximum selectivity of 85%. Both cumene and ethylbenzene oxidation are performed without a catalyst. The world production of cumene hydroperoxide is estimated at 17 million tons/year, and production of ethylbenzene hydroperoxide is estimated as 6 million tons/year [1]. On a smaller scale, hydroperoxides are used as semi-products in synthesis of resorcinol, hydroquinone and cresols via methods analogous to phenol synthesis [1].

Potentially, oxidation of isopropylaromatics to hydroperoxides may additionally constitute an environment-friendly alternative for synthesis of other valuable hydroxyaromatic compounds, which are still produced by classical stoichiometric methods, e.g., 2-naphthol from naphthalene by sulfonation [3] or 4,4'-dihydroxybiphenyl by oxidative coupling of 2,6-di-*tert*-butylphenol [4], which are characterized by a low atom economy and a high environmental factor.

Because of the significant role of oxidation of alkylaromatic hydrocarbons in the industry, new methods are required that would enable improved technological and economic indicators. Remarkable progress was made in this field by the use of *N*-hydroxyimides, such as *N*-hydroxyphthalimide (NHPI), as catalysts [5–9]; in mild conditions, not only high yields but also high selectivity to hydroperoxides were obtained. The activity of NHPI results from the formation of 1,3-dioxoindole-2-oxyl radical (PINO) in the system, which effectively abstracts a hydrogen atom from oxidized hydrocarbon (Fig. 2), with a higher rate than peroxy radicals in autocatalytic processes. In general terms, PINO from NHPI in the oxidation of hydrocarbons is most commonly generated by additives such as transition metal salts, primarily cobalt(II) salts, azo compounds, aldehydes, benzoquinone, anthraquinone and dimethylglyoxime [5–9]. However, in the case of hydroperoxide synthesis, the use of metal salts is limited due to their catalytic influence on the decomposition of hydroperoxides.

The oxidation of hydrocarbons to hydroperoxides with use of NHPI has been previously described, such as that of cumene [10–14], ethylbenzene [14–16], *p*-cymene [10,14], 1,4-dimethyl-2-isopropylbenzene,

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